

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12387504
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Matsumoto; Yuki et al.

Image processing apparatus

Abstract

An image processing apparatus includes a controller configured to: (i) perform image recognition of a camera image to identify a signal region in which a traffic light exists in the camera image, the traffic light including an arrow light; (ii) identify a candidate region of the signal region, the candidate region satisfying (a) a first condition related to a color component of the arrow light and (b) a second condition related to a luminance component of the arrow light; and (iii) determine that the arrow light of the traffic light is turned on when a size of the candidate region that has been identified is smaller than a predetermined size.

Inventors: Matsumoto; Yuki (Kobe, JP), Kakita; Naoshi (Kobe, JP), Kono; Takashi (Kobe, JP), Kinoshita; Shota (Kobe, JP), Nishijima; Yasutaka (Kobe, JP)

Applicant: DENSO TEN Limited (Kobe, JP)

Family ID: 92714539

Assignee: DENSO TEN Limited (Kobe, JP)

Appl. No.: 18/583975

Filed: February 22, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240312224 A1	Sep. 19, 2024

Foreign Application Priority Data

JP	2023-040917	Mar. 15, 2023
----	-------------	---------------

Publication Classification

Int. Cl.: G06V20/58 (20220101); G06T7/11 (20170101)

U.S. Cl.:

CPC G06V20/584 (20220101); G06T7/11 (20170101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2015/0186734	12/2014	Nakagawa	382/104	G06V 20/584
2017/0228606	12/2016	Guan	N/A	G06T 7/90
2023/0134330	12/2022	Horiba	382/106	G06T 7/50
2024/0208501	12/2023	Kitahara	N/A	B60W 50/14
2024/0404300	12/2023	Ikeda	N/A	G06V 20/584

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2022-108760	12/2021	JP	N/A

Primary Examiner: Monshi; Samira

Attorney, Agent or Firm: Oliff PLC

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The invention relates to an image processing apparatus, an image processing method and a non-transitory computer-readable recording medium.

Description of the Background Art

(2) Conventionally, in a case where an arrow light of a traffic light is detected from a camera image of a vehicle, a technology that estimates a position of a red light based on the detected arrow light, and when the red light is detected in an estimation position, detects the arrow light as a true arrow light has been proposed (for example, refer to the Japanese Published Unexamined Patent Application No. 2022-108760).

(3) However, when the above conventional technology is used, a position of a pixel that is detected as the red light due to halation at night is deviated so that the arrow light may be undetected although the arrow light is actually turned on.

(4) Furthermore, when the red light of the traffic light without the arrow light is turned on, if there happens to be a background, such as a tree having the same color component as that of the arrow light, in a position of the arrow light in the traffic light with the arrow light, the arrow light may be erroneously detected.

SUMMARY OF THE INVENTION

- (5) According to one aspect of the invention, an image processing apparatus includes a controller configured to: (i) perform image recognition of a camera image to identify a signal region in which a traffic light exists in the camera image, the traffic light including an arrow light; (ii) identify a candidate region of the signal region, the candidate region satisfying (a) a first condition related to a color component of the arrow light and (b) a second condition related to a luminance component of the arrow light; and (iii) determine that the arrow light of the traffic light is turned on when a size of the candidate region that has been identified is smaller than a predetermined size.
- (6) It is an object of the invention to provide an image processing apparatus, an image processing method and a non-transitory computer-readable recording medium capable of improving a detection accuracy of an arrow light in a traffic light.
- (7) These and other objects, features, aspects and advantages of the invention will become more apparent from the following detailed description of the invention when taken in conjunction with the accompanying drawings.
-

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic explanatory diagram (No. 1) of an image processing method according to an embodiment;
- (2) FIG. 2 is an explanatory diagram in a case where a conventional arrow light is undetected or erroneously detected;
- (3) FIG. 3 is a schematic explanatory diagram (No. 2) of the image processing method according to the embodiment;
- (4) FIG. 4 illustrates a configuration example of an image processing apparatus according to the embodiment;
- (5) FIG. 5 illustrates a configuration example of an image processing apparatus according to an example modification;
- (6) FIG. 6 is a flowchart (No. 1) illustrating a processing procedure executed by the image processing apparatus according to the embodiment;
- (7) FIG. 7 is a flowchart (No. 2) illustrating a processing procedure executed by the image processing apparatus according to the embodiment;
- (8) FIG. 8 is a flowchart (No. 3) illustrating a processing procedure executed by the image processing apparatus according to the embodiment;
- (9) FIG. 9 is an explanatory diagram illustrating a conventional case of undetected or erroneous detection improved by the embodiment; and
- (10) FIG. 10 is an explanatory diagram in a case where a signal region is rotated 90 degrees.

DESCRIPTION OF THE EMBODIMENTS

- (11) An embodiment of an image processing apparatus, an image processing method, and a non-transitory computer-readable recording medium disclosed in the present application will be described in detail below with reference to the accompanying drawings. This invention is not limited to the embodiment described below.
- (12) In the following, it is assumed that an image processing apparatus **10** according to the embodiment is an in-vehicle apparatus to be mounted in a vehicle. The image processing apparatus **10** is provided to detect an object and a state of the object in a camera image by image recognition of the camera image. In this embodiment, it is assumed that the object to be detected is a traffic light **300**. Furthermore, it is assumed that the state of the object to be detected is a lighting state of the traffic light **300**.
- (13) Hereinafter, the expression “prescribed” or “certain” may be replaced with “predetermined”.
- (14) First, an outline of the image processing method according to the embodiment will be

described with reference to FIG. 1 to FIG. 3. FIG. 1 is a schematic explanatory diagram (No. 1) of the image processing method according to the embodiment. FIG. 2 is an explanatory diagram in a case where a conventional arrow light is undetected or erroneously detected. FIG. 3 is a schematic explanatory diagram (No. 2) of the image processing method according to the embodiment.

(15) The image processing apparatus **10** detects the lighting state of the traffic light **300**, and for example, determines whether or not a driver of the vehicle is ignoring the traffic light based on the detection result. Such information processing is executed by a controller **12** (refer to FIG. 4) included in the image processing apparatus **10**.

(16) As illustrated in FIG. 1, the controller **12** inputs the camera image and identifies a rectangular region in which the traffic light **300** exists from the image. (a step S1). In the step S1, an existing image processing technology is applied. In the step S1, the controller **12** identifies the rectangular region in which the traffic light **300** exists from the image using a DNN (Deep Neural Network) model, etc., learned by using a machine learning algorithm. In the following, such a rectangular region is referred to as a “signal region SR”.

(17) Subsequently, the controller **12** inputs the signal region SR to determine the lighting state of the traffic light **300** (a step S2). In the step S2, the controller **12** analyzes the image of the signal region SR to determine whether the traffic light **300** is in a green lighting state, a yellow lighting state, or a red lighting state. In the red lighting state, the controller **12** further determines whether or not an arrow light of the traffic light **300** is turned on.

(18) Next, the controller **12** performs motion estimation of estimating how the traffic light **300** moves relative to a traveling vehicle by a difference between frames of the camera image (a step S3). The controller **12** determines whether or not the driver of the vehicle is ignoring the traffic light based on a result of a lighting state determination in the step S2 and a result of the motion estimation in the step S3 (a step S4), and outputs the determination result.

(19) The image processing method according to the embodiment is applied in the step S2 surrounded by a dashed rectangle. In the conventional technology, in a case where the arrow light of the traffic light **300** is detected in the step S2, the arrow light is detected based on a positional relationship with a red light.

(20) However, when the conventional technology is used, the arrow light may be undetected although the arrow light is actually turned on, or the arrow light may be erroneously detected although the traffic light **300** does not include the arrow light.

(21) Specifically, a case in which the arrow light is undetected is, as illustrated in a left drawing of FIG. 2, a case in which a center position RP of the red light is erroneously detected, for example, due to halation at night. In this case, if the controller **12** tries to detect the arrow light based on the positional relationship between the red light and arrow light, the arrow light does not exist in a region P1 in which the arrow light should exist relative to the center position RP, so that the controller **12** determines that the arrow light is undetected.

(22) A case in which the arrow light is erroneously detected is, as illustrated in a right drawing of FIG. 2, for example, a case in which there is a background, such as a tree T, in the region P1 relative to the center position RP, although the center position RP of the red light has been normally detected. In this case, if the background happens to have a same color component as that of the arrow light, the controller **12** erroneously detects the arrow light.

(23) Thus, in the image processing method according to the embodiment, when the controller **12** detects the arrow light in the step S2, the controller **12** identifies a candidate region that satisfies a first condition related to the color component of the arrow light and a second condition related to a luminance component of the arrow light from the signal region SR. When a size of the identified candidate region is smaller than a predetermined size, the controller **12** determines that the arrow light of the traffic light **300** is turned on.

(24) Specifically, as illustrated in FIG. 3, in the image processing method according to the embodiment, the controller **12** divides the identified signal region SR into a plurality of sub-regions

(a step S21). FIG. 3 shows an example in which the signal region SR is divided into 16 ($=4 \times 4$) sub-regions. However, a number of divisions of the signal region SR is not limited thereto.

(25) Subsequently, the controller **12** identifies the plurality of the sub-regions that satisfy the following two conditions (a step S22). The first condition as one of the two conditions is a condition related to the color component of the arrow light. Specifically, as illustrated in FIG. 3, the first condition is that a size of a pixel region having the color component of the arrow light that is turned on is a predetermined value or more.

(26) The second condition as the other of the two conditions is a condition related to the luminance component of the arrow light. Specifically, as illustrated in FIG. 3, the second condition is that a number of pixels with luminance values above a certain level is the predetermined value or more. The controller **12** identifies the plurality of the sub-regions that satisfy both the first and second conditions as the candidate region.

(27) In this way, the controller **12** divides the signal region SR into the plurality of the sub-regions and identifies the candidate region corresponding to the arrow light for each of the plurality of the sub-regions. Thus, it is possible to easily determine whether or not the arrow light is turned on by a number of the identified candidate regions.

(28) When the number of the identified candidate regions is less than a predetermined number in a vertical direction, the controller **12** determines that the arrow light of the traffic light **300** is turned on (a step S23). The predetermined number is set to a value (e.g., three) such as a threshold value converted to the number of the plurality of the sub-regions at which a size of the candidate region that satisfies the conditions corresponding to the arrow light is considered to be too large. However, the predetermined number is not limited thereto, and an arbitrary value may be set.

(29) Here, although an example in which the number of the identified candidate regions is determined has been described, the number of the identified candidate regions is converted to the size of the candidate region. That is, when the size of the identified candidate region is smaller than the predetermined size, the controller **12** determines that the arrow light of the traffic light **300** is turned on. This method of determining the size of the candidate region is used regardless of a presence or absence of the plurality of the sub-regions. Therefore, the signal region SR does not always have to be divided.

(30) As illustrated in FIG. 3, in the image processing method according to the embodiment, when the controller **12** detects the arrow light in the step S2, the controller identifies the candidate region that satisfies the first condition related to the color component of the arrow light and the second condition related to the luminance component of the arrow light from the signal region SR. Then, when the size of the identified candidate region is smaller than the predetermined size, the controller **12** determines that the arrow light of the traffic light **300** is turned on.

(31) As described above, since the controller **12** performs determination of the arrow light without depending on the positional relationship between the red light and the arrow light, it is possible to suppress a case of undetected or erroneous detection illustrated in FIG. 2. That is, according to the image processing method according to the embodiment, it is possible to improve a detection accuracy of the arrow light in the traffic light **300**.

(32) In the explanation of FIG. 3, although, in the step S23, the controller **12** determines the signal region SR in the vertical direction, this is because the traffic light **300** in the signal region SR is horizontally oriented. When the traffic light **300** in the signal region SR is vertically oriented, in the step S23, the controller **12** may determine the signal region SR in a horizontal direction.

(33) When the traffic light **300** is vertically oriented, the controller **12** may rotate the signal region SR 90 degrees so that the traffic light **300** is regarded as being horizontally oriented. This point will be described later with reference to FIG. 10.

(34) Hereinafter, a configuration example of the image processing apparatus **10** using the image processing method according to the embodiment will be more specifically described.

(35) FIG. 4 illustrates the configuration example of the image processing apparatus **10** according to

the embodiment. As illustrated in FIG. 4, the image processing apparatus 10 includes a memory 11 and the controller 12. A camera 3 and an output portion 5 are connected to the image processing apparatus 10.

(36) The camera 3 is mounted in the vehicle and provided to capture an image in front of the vehicle. The camera 3 may be a 360-degree camera capable of capturing images not only in front of the vehicle but also all around the vehicle.

(37) The output portion 5 is an output device that presents output information from the image processing apparatus 10. The output portion 5 is implemented by a display, a speaker, and the like.

(38) The image processing apparatus 10 is, for example, a computer to be mounted in the vehicle. The image processing apparatus 10 performs an information processing of at least the steps S1 to S4 described with reference to FIG. 1 and the step S21 to S23 described with reference to FIG. 3.

(39) The memory 11 is implemented by a storage device such as a RAM (Random Access Memory) or a flash memory. The memory 11 stores a computer program according to the embodiment executed by the controller 12. Furthermore, the memory 11 stores various information that is used in the information processing executed by the controller 12.

(40) The memory 11 stores, for example, an AI (artificial Intelligence) model for the image recognition. The memory 11 stores various parameter information including the first and second conditions described above.

(41) The controller 12 corresponds to a processor. The controller 12 is implemented by a CPU (Central Processing Unit), an MPU (Micro Processing Unit), a GPU (Graphics Processing Unit), or the like. The controller 12 reads and executes a computer program according to the embodiment stored in the memory 11 using the RAM as a work area. The controller 12 is also implemented by an integrated circuit such as an ASIC (Application Specific Integrated Circuit) or a FPGA (Field Programmable Gate Array).

(42) The controller 12 performs the information processing according to the processing procedure illustrated in each flowchart of FIG. 6 to FIG. 8. An explanation with reference to FIG. 6 to FIG. 8 will be described later.

(43) As illustrated in FIG. 4, the camera 3, the output portion 5 and the image processing apparatus 10 are implemented as a drive recorder 1. In this case, the camera 3 is implemented by a camera unit to be mounted in the drive recorder 1. The output portion 5 is implemented by a display and a speaker to be mounted in the drive recorder 1. The image processing apparatus 10 is implemented by a microcomputer to be mounted in the drive recorder 1.

(44) The configuration example illustrated in FIG. 4 is one example, and besides this, a modification example will be given. FIG. 5 illustrates the configuration example of the image processing apparatus 10 according to an example modification. As illustrated in FIG. 5, the image processing apparatus 10 is implemented as an image processing ECU (Electronic Control Unit) 9. In this case, the camera 3 is, for example, implemented by the camera unit to be mounted in the drive recorder 1. The camera 3 may be implemented by an in-vehicle camera unit other than the drive recorder 1. The output portion 5 is implemented by an in-vehicle display and an in-vehicle speaker as an in-vehicle output device 7 to be mounted in the vehicle.

(45) In the example of FIG. 5, the camera 3 and the output portion 5 are connected to the image processing apparatus 10 via an in-vehicle network, such as a CAN (Controller Area Network). The camera 3 and the output portion 5 may be connected to the image processing apparatus 10 via a Bluetooth (registered trademark), a Wi-Fi (registered trademark), a UWB (Ultra-Wide Band), or the like, by using a wireless network.

(46) Next, a processing procedure of the information processing executed by the controller 12 of the image processing apparatus 10 will be described with reference to FIG. 6 and FIG. 8. FIG. 6 is a flowchart (No. 1) illustrating the processing procedure executed by the image processing apparatus 10 according to the embodiment. FIG. 7 is a flowchart (No. 2) illustrating the processing procedure executed by the image processing apparatus 10 according to the embodiment. FIG. 8 is a

flowchart (No. 3) illustrating the processing procedure executed by the image processing apparatus **10** according to the embodiment.

(47) In order to determine whether or not the driver of the vehicle is ignoring the traffic light as illustrated in FIG. **1**, the controller **12** acquires the camera image from the camera **3** as illustrated in FIG. **6** (a step **S101**). Subsequently, the controller **12** identifies the signal region SR in which the traffic light **300** exists from the acquired camera image (a step **S102**).

(48) Subsequently, the controller **12** performs a “lighting state determination process” that determines the lighting state of the traffic light **300** based on the identified signal region SR (a step **S103**).

(49) In this “lighting state determination process”, as illustrated in FIG. **7**, the controller **12** determines whether the traffic light **300** is in the green lighting state, the yellow lighting state, or the red lighting state by image analysis of the identified signal region SR (a step **S201**).

(50) In the step **S201**, the controller **12**, for example, applies a color extraction filter that extracts green, yellow, and red components from the signal region SR. The controller **12** determines whether the traffic light **300** is in the green lighting state, the yellow lighting state, or the red lighting state, for example, based on a color corresponding to the color component with the highest amount out of the extracted color components.

(51) Subsequently, the controller **12** performs a “lighting state determination process of the arrow light” (a step **S202**). Although the figure is omitted, the step **S202** may be performed when the traffic light **300** has been determined to be in the red lighting state in the step **S201**.

(52) In the “lighting state determination process of the arrow light”, as illustrated in FIG. **8**, the controller **12** divides the signal region SR into the plurality of the sub-regions (a step **S301**). The controller **12** divides the signal region SR into the plurality of the sub-regions each having a predetermined size.

(53) Subsequently, the controller **12** selects one of the plurality of the sub-regions (a step **S302**). Then, the controller **12** determines whether or not the selected one sub-region satisfies the condition related to the color component (i.e., the above first condition) (a step **S303**).

(54) When the selected one sub-region satisfies the condition related to the color component (Yes in the step **S303**), then, the controller **12** determines whether or not the selected one sub-region satisfies the condition related to the luminance component (i.e., the above second condition) (a step **S304**).

(55) When the selected one sub-region satisfies the condition related to the luminance component (Yes in the step **S304**), the controller **12** identifies the selected one sub-region as the candidate region that satisfies the condition (a step **S305**).

(56) On the other hand, when the selected one sub-region does not satisfy the condition related to the color component (No in the step **S303**) or does not satisfy the condition related to the luminance component (No in the step **S304**), the controller **12** does not identify the selected one sub-region as the candidate region that satisfies the conditions (a step **S306**).

(57) The controller **12** determines whether or not there is an undetermined sub-region (a step **S307**). When there is the undetermined sub-region (Yes in the step **S307**), the controller **12** repeats the process from the step **S302**. When there is no undetermined small area (No in the step **S307**), the controller **12** determines whether or not there is an identified candidate region (a step **S308**).

(58) When there is the identified candidate region (Yes in the step **S308**), the controller **12** determines whether or not the number of the identified candidate regions is less than the predetermined number in the vertical direction (a step **S309**). When the number of the identified candidate regions is less than the predetermined number (Yes in the step **S309**), the controller **12** determines that the arrow light of the traffic light **300** is turned on (a step **S310**).

(59) When there is no identified candidate region (No in the step **S308**), or the number of the identified candidate regions is the predetermined number or more in the vertical direction (No in the step **S309**), the controller **12** determines that the arrow light of the traffic light **300** is not turned

on (a step S311). Then, the controller **12** terminates the lighting state determination process of the arrow light.

(60) Returning back to FIG. 7, when the controller **12** terminates the step S202, the controller **12** returns a result of the lighting state determination process (a step S203). The controller **12** terminates the lighting state determination process.

(61) Returning back to FIG. 6, when the controller **12** terminates the step S103, the controller **12** performs the motion estimation of the traffic light **300** relative to the vehicle (a step S104). The controller **12** estimates how the traffic light **300** moves relative to the traveling vehicle, as described above, for example, by the difference between the frames of the camera image.

(62) Subsequently, the controller **12** determines whether or not the driver of the vehicle is ignoring the traffic light based on the result of the lighting state determination process in the step S103 and the result of the motion estimation in the step S104 (a step S105). Then, the controller **12** outputs the determination result to the output portion **5** (a step S106), and terminates the process.

(63) Although the figure is omitted, the controller **12** may output the determination result to the output portion **5** based on a stability of the determination result between the frames of the camera image. In one example, when the stability is considered to be high, such a case that the same determination result is consecutive between frames, the controller **12** may output the determination result to the output portion **5**. As a result, it is possible to output the stable determination result that is hardly affected by disturbances.

(64) According to the image processing apparatus **10** according to the embodiment described above, the conventional case of undetected or erroneous detection described with reference to FIG. 2 is improved. FIG. 9 is an explanatory diagram illustrating the conventional case of undetected or erroneous detection improved by the embodiment.

(65) In the explanation of FIG. 9, it is assumed that the predetermined number for determining the number of the identified candidate regions is three. As illustrated in a left drawing of FIG. 9, in the conventional case in which the arrow light is undetected, according to the embodiment, for example, even when there is halation H, if the first and the second conditions of the arrow light are satisfied, the candidate region of the arrow light is identified. When the number of the identified candidate regions is less than three in a vertical direction CD, the arrow light is detected. That is, according to the embodiment, even when there is the halation H, the arrow light is correctly determined.

(66) On the other hand, as illustrated in a right drawing of FIG. 9, in the conventional case in which the arrow light is erroneously detected, according to the embodiment, for example, even when there is a tree T having the same color component as that of the arrow light, if the number of the identified candidate regions is three or more in the vertical direction CD, the arrow light is not detected. That is, according to the embodiment, it is possible to correctly determine that the background having the same color component as that of the arrow light is not the arrow light.

(67) The right drawing of FIG. 9 illustrates the traffic light **300** without the arrow light. However, in a case where the traffic light **300** includes the arrow light, it is expected that the background having the same color component as that of the arrow light is reflected in the camera image.

(68) In this case, in order to distinguish the color component of the arrow light from the color component corresponding to the background of the traffic light **300**, such a countermeasure that the threshold value for distinguishing between both the color components is set for the above first condition is taken. As a result, when the traffic light **300** includes the arrow light, even when the background having the same color component as that of the arrow light is reflected in the camera image, it is possible to correctly determine the arrow light.

(69) Furthermore, as illustrated in the right drawing of FIG. 9, when the background is the tree T, etc., the candidate regions are often consecutively identified in the vertical direction CD of the signal region SR. Therefore, when there are consecutive ones of the candidate regions in the vertical direction CD of the signal region SR, the controller **12** may determine that the arrow light

of the traffic light **300** is not turned on without depending on the number of the candidate regions. As a result, the controller **12** correctly determines that the arrow light of the traffic light **300** is not turned on based on such a feature that the consecutive candidate regions are likely to be identified in a case where the background is the tree T, etc.

(70) By the way, in the explanation of FIG. **9**, an example in which, when the traffic light **300** in the signal region SR is horizontally oriented, the controller **12** determines the signal region SR in the vertical direction CD has been described. On the other hand, when the traffic light **300** in the signal region SR is vertically oriented, the controller **12** may determine the signal region SR in the horizontal direction. That is, the controller **12** determines the signal region SR in the same direction of either the vertical direction or the horizontal direction, depending on whether the traffic light **300** in the signal region SR is horizontally or vertically oriented. As a result, regardless of whether the traffic light **300** is horizontally or vertically oriented, it is possible to detect the arrow light in a similar determination method.

(71) Furthermore, when the traffic light **300** is vertically oriented, the controller **12** may rotate the signal region SR 90 degrees so that the traffic light **300** is regarded as being horizontally oriented. FIG. **10** is an explanatory diagram in a case where the signal region SR is rotated 90 degrees.

(72) The controller **12** samples images of the traffic light **300** having various sizes that are reflected in the camera image so that the images of the traffic light **300** always have the same size (e.g., 25×25 pixels) so as to identify the signal region SR.

(73) For a sampling direction, as illustrated in FIG. **10**, a horizontal traffic light is scanned from the top left to the bottom right and a vertical traffic light is scanned from the bottom left to the top right. That is, the controller **12** samples the images of the traffic light **300** in the sampling direction in which the pixels on the camera image are sequentially scanned along a longitudinal direction of the traffic light **300** in the camera image so as to identify the signal region SR. Thus, as illustrated in FIG. **10**, the signal region SR after sampling is seen the same regardless of whether the traffic light **300** is horizontally or vertically oriented.

(74) Therefore, for the traffic light **300** to be determined, the controller **12** treats the images of the traffic light **300** as image information that always has the same size and direction, regardless of the size of the traffic light **300** in the camera image and whether the traffic light **300** is horizontally or vertically oriented. In this case, the controller **12** determines the identified candidate region always in the vertical direction CD.

(75) As described above, the image processing apparatus **10** according to the embodiment has the controller **12**. The controller **12** performs image recognition of the camera image to identify the signal region SR in which the traffic light **300** exists in the camera image. The controller **12** identifies the candidate region that satisfies the first condition related to the color component of the arrow light and the second condition related to the luminance component of the arrow light from the signal region SR. When the size of the identified candidate region is smaller than the predetermined size, the controller **12** determines that the arrow light of the traffic light **300** is turned on. As a result, it is possible to prevent the arrow light from being undetected although the arrow light is actually turned on or to prevent the arrow light from being erroneously detected in a case where the traffic light **300** does not include the arrow light. That is, according to the image processing apparatus **10** according to the embodiment, it is possible to improve the detection accuracy of the arrow light in the traffic light **300**.

(76) In the embodiment described above, although the lighting state determination and motion estimation of the traffic light **300** and determination of whether or not the driver of the vehicle is ignoring the traffic light are performed based on the image recognition, sensor data of various sensors mounted in the vehicle may be naturally combined with the image recognition. For example, a behavior of the vehicle may be estimated by using a sensor value of a steering sensor or an acceleration sensor. An own vehicle speed may be acquired by using a sensor value of a speed sensor.

(77) In the embodiment described above, although the signal region SR is a rectangular region, the signal region SR does not have to be a rectangular region.

(78) It is possible for a person skilled in the art to easily come up with more effects and modifications. Thus, a broader modification of this invention is not limited to specific description and typical embodiments described and expressed above. Therefore, various modifications are possible without departing from the general spirit and scope of the invention defined by claims attached and equivalents thereof.

(79) While the invention has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore understood that numerous other modifications and variations can be devised without departing from the scope of the invention.

Claims

1. An image processing apparatus comprising a controller configured to: (i) perform image recognition of a camera image to identify a signal region in which a traffic light exists in the camera image, the traffic light including an arrow light; (ii) identify a candidate region of the signal region, the candidate region satisfying (a) a first condition related to a color component of the arrow light and (b) a second condition related to a luminance component of the arrow light; and (iii) determine that the arrow light of the traffic light is turned on when a size of the candidate region that has been identified is smaller than a predetermined size.
2. The image processing apparatus according to claim 1, wherein (1) the signal region is a rectangular region, (2) the rectangular region is divided into a plurality of sub-regions, (3) a determination is made as to whether the first and second conditions are satisfied for each of the plurality of sub-regions, and (4) when one or more of the plurality of sub-regions satisfy the first and second conditions, the one or more of the sub-regions are identified as the candidate regions.
3. The image processing apparatus according to claim 2, wherein when a number of the candidate regions that have been identified is less than a predetermined number in a same direction as a longer direction of the rectangular region, it is determined that the arrow light of the traffic light is turned on.
4. The image processing apparatus according to claim 2, wherein when there are consecutive ones of the candidate regions arranged in a same direction as a longer direction of the rectangular region, it is determined that the arrow light of the traffic light is not turned on.
5. The image processing apparatus according to claim 3, wherein when the traffic light is vertically oriented, the signal region is rotated 90 degrees so that the traffic light is regarded as being horizontally oriented.
6. The image processing apparatus according to claim 5, wherein when the number of the candidate regions that have been identified is less than the predetermined number in a vertical direction of the rectangular region, it is determined that the arrow light of the traffic light is turned on.
7. The image processing apparatus according to claim 1, wherein the candidate region is identified based on the first condition provided to distinguish the color component of the arrow light from a color component corresponding to a background of the traffic light.
8. The image processing apparatus according to claim 1, wherein a determination result of a lighting state of the traffic light is output based on a stability of the determination result between frames of the camera image.
9. The image processing apparatus according to claim 1, wherein the image processing apparatus is mounted in a drive recorder.
10. An image processing method executed by an image processing apparatus, the method comprising the steps of: (a) performing image recognition of a camera image to identify a signal region in which a traffic light exists in the camera image, the traffic light including an arrow light; (b) identifying a candidate region of the signal region, the candidate region satisfying (i) a first

condition related to a color component of the arrow light and (ii) a second condition related to a luminance component of the arrow light; and (c) determining that the arrow light of the traffic light is turned on when a size of the candidate region that has been identified is smaller than a predetermined size.

11. The image processing method according to claim 10, the method further comprising the steps of: (d) identifying the signal region as a rectangular region; (e) dividing the rectangular region into a plurality of sub-regions; (f) determining whether the first and second conditions are satisfied for each of the plurality of sub-regions; and (g) when one or more of the plurality of sub-regions satisfy the first and second conditions, identifying the one or more sub-regions as the candidate regions.

12. A non-transitory computer-readable recording medium having stored therein a program that causes a computer of an image processing apparatus to execute a process, the process comprising: (i) performing image recognition of a camera image to identify a signal region in which a traffic light exists in the camera image, the traffic light including an arrow light; (ii) identifying a candidate region of the signal region, the candidate region satisfying (a) a first condition related to a color component of the arrow light and (b) a second condition related to a luminance component of the arrow light; and (iii) determining that the arrow light of the traffic light is turned on when a size of the candidate region that has been identified is smaller than a predetermined size.

13. The non-transitory computer-readable recording medium according to claim 12, the process further comprising: (iv) identifying the signal region as a rectangular region; (v) dividing the rectangular region into a plurality of sub-regions; (vi) determining whether the first and second conditions are satisfied for each of the plurality of sub-regions; and (vii) when one or more of the plurality of sub-regions satisfy the first and second conditions, identifying the one or more sub-regions as the candidate regions.
